

Genome to Phenome Mapping in Apple Using Historical Data

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Abstract

Apple (*Malus X. domestica* Borkh.) is one of the world's most valuable fruit crops. Its large size and long juvenile phase make it a particularly promising candidate for marker-assisted selection (MAS). However, advances in MAS in apple have been limited by a lack of phenotype and genotype data from sufficiently large samples. To establish genotype-phenotype relationships and advance MAS in apple, we extracted over 24,000 phenotype scores from the USDA-Germplasm Resources Information Network (GRIN) database and linked them with over 8000 single nucleotide polymorphisms (SNPs) from 689 apple accessions from the USDA apple germplasm collection clonally preserved in Geneva, NY. We find significant genetic differentiation between Old World and New World cultivars and demonstrate that the genetic structure of the domesticated apple also reflects the time required for ripening. A genome-wide association study (GWAS) of 36 phenotypes confirms the association between fruit color and the MYB1 locus, and we also report a novel association between the transcription factor, NAC18.1, and harvest date and fruit firmness. We demonstrate that harvest time and fruit size can be predicted with relatively high accuracies ($r > 0.46$) using genomic prediction. Rapid decay of linkage disequilibrium (LD) in apples means millions of SNPs may be required for well-powered GWAS. However, rapid LD decay also promises to enable extremely high resolution mapping of causal variants, which holds great potential for advancing MAS.

TO MEET THE NEEDS of a growing global population, food availability must double within 25 yr (McCouch et al., 2013). Fortunately, advances in genomics allow breeders to more accurately and quickly improve crops (Lusser et al., 2012). However, continued food improvement relies on increasing our understanding of the genome-phenome relationship (Morrell et al., 2011; Varshney et al., 2014). Association mapping can detect causal genes of interest using phenotyped populations of unrelated individuals, such as those already available in germplasm collections. Once genetic markers linked to important traits are discovered, marker-assisted breeding can enable more efficient selection for plants with these desirable characteristics (Morrell et al., 2011; McCouch et al., 2013).

Apple had the third highest global gross production value among fruit crops in 2013, and is well-poised to benefit from MAS that would eliminate undesirable genotypes at the early seedling stage (Myles, 2013; McClure et al., 2014; Food and Agriculture Organization

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Abbreviations: AIC, Akaike Information Criterion; GBS, genotyping-by-sequencing; GDR, Genome Database for Rosaceae; GRIN, Germplasm Resources Information Network; GWAS, genome-wide association study; IBD, identity-by-descent; LD, linkage disequilibrium; MAS, marker-assisted selection; OR, odds-ratios; PC, principal component; PCA, principal components analysis; SNP, single nucleotide polymorphism.

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of the United Nations, 2015). Apples have a long juvenile period: significant fruiting generally occurs 5 yr after germination, even when using a quickly maturing dwarf rootstock (Kumar et al., 2012). Two to three additional years may be required to phenotype fruit quality traits before selecting parents for crosses, and a large percentage of offspring are discarded within the first decade of fruit evaluation (Kumar et al., 2013b; Myles, 2013). Apple breeding is also limited by self-incompatibility and a large tree size that requires substantial space and money to breed (Brown and Maloney, 2003; Myles, 2013; McClure et al., 2014). As a result, one recent breeding program required 26 yr to generate three commercial cultivars from a starting population of 52,000 seedlings (Peil et al., 2008).

Markers linked to numerous phenotypes have been discovered in apple, and MAS is already being applied for traits such as disease resistance, postharvest storability, skin color, and fruit texture, as well as dwarfing and precocity in rootstocks (Fazio et al., 2014; Ru et al., 2015). In one example, using MAS to select for a single marker linked to postharvest storability resulted in an estimated savings of at least 60% of the operating costs associated with first-stage seedling selection (Edge-Garza et al., 2010).

A major barrier to establishing robust genotype-phenotype relationships that can be leveraged for MAS is poor quality phenotype data (Benfey and Mitchell-Olds, 2008; Cobb et al., 2013; Meneses and Orellana, 2013). While technological advances continue to increase the speed and decrease the cost of acquiring genetic data, slow and expensive phenotyping results in a “phenotyping bottleneck” (Houle et al., 2010; Kumar et al., 2012; Burleigh et al., 2013). It is well known that high-quality phenotype data often results in far better powered quantitative trait locus (QTL) analyses (Van Eerdewegh et al., 2002). Fortunately, improvements to phenotyping technology have begun and the scientific community generally recognizes the need for high-quality phenotypic measures (Houle et al., 2010; Furbank and Tester, 2011; Meneses and Orellana, 2013; Deans et al., 2015).

There has been great support for the use of historical phenotype data from gene banks for genetic mapping. However, phenotypic evaluation is especially challenging and costly over long time periods, and using data not collected specifically for genetic mapping is often problematic (Myles et al., 2009; Houle et al., 2010; McCouch et al., 2012). Different observers measuring traits over multiple years in varying environments cause phenotyping discrepancies in historical data sets. While DNA sequences are comparable between studies, phenotype data are much more difficult to compare because of missing data, inconsistent replication, and the frequent use of nonquantitative measurements (Peace and Norelli, 2009; Houle et al., 2010; McCouch et al., 2012).

Despite the difficulty of acquiring reliable data and the subsequent need for curation, historical phenotype data has been successfully employed to identify genotype-phenotype relationships in barley (*Hordeum vulgare*

L.) and potato (*Solanum tuberosum* L.) (Baldwin et al., 2011; Matthies et al., 2014). Here, we examine historical phenotype data available from a large apple gene bank, the USDA apple germplasm collection, and link them to genotypes collected using genotyping-by-sequencing (GBS) (Elshire et al., 2011). We find relationships of interest between phenotypes, identify several genotype-phenotype associations using GWAS, describe the very rapid LD decay in the domesticated apple, and quantify our ability to predict phenotypes using genomic prediction.

Materials and Methods

Phenotype Scoring and Filtering

Publicly available phenotype data were downloaded from the USDA-GRIN website (https://npgsweb.ars-grin.gov/gringlobal/view2.aspx?dv=web_taxonomy&species_view_accessionlist¶ms=:taxonomyid=104681; accessed 18 July 2011; verified 24 Mar. 2016). A description of the steps to edit and curate the phenotype data is provided in Supplemental Fig. S1. Phenotype data were first trimmed to exclude accessions not labeled as *Malus × domestica* Borkh., as well as outliers that were clearly mislabeled and did not fall within the *M. × domestica* variation observed according to a principal components analysis (PCA) of the genetic data. We manually curated and recoded phenotypes when phenotype scoring was incompatible with downstream applications (Supplemental Table S1). In several cases, such as recoding color as a binary trait, data points were removed as a result. Using the genetic data, we determined which accessions exhibited clonal relationships and measurements across clones were averaged (see *Genetic Analyses* below). Phenotypes were also combined across years and averaged in cases of replication for a particular accession. Categorical phenotypes were excluded from the analysis. We removed phenotypes containing data for fewer than 100 accessions, as well as invariable phenotypes scored entirely as one value. Binary phenotypes with highly uneven distributions of trait scores (i.e., one of the two values was present at a frequency > 95%) were also excluded. The final phenotype data set included binary, ordinal, and quantitative phenotypes.

Associations between phenotypes were tested using Pearson's correlation for binary-binary, quantitative-quantitative, and binary-quantitative comparisons. Spearman's rank correlation was used for binary-ordinal and quantitative-ordinal comparisons, while Kendall's rank correlation was used for ordinal-ordinal comparisons. Performing correlations between every possible pair of phenotypes generated a pairwise correlation matrix. To correct for multiple comparisons, a Bonferroni correction was applied by multiplying *p*-values by the number of pairwise comparisons (630).

We divided accessions into several binary categories including harvest season (early and late), color (red and green/yellow), use (cider and eating/cooking) and origin (New World and Old World) using information from the USDA-GRIN database when possible and online sources

otherwise (Supplemental Table S2). We tested whether phenotypes showed differences according to these categories using a Fisher's exact test for binary phenotypes and a Mann–Whitney U test for ordinal and quantitative phenotypes. For Fisher's exact test, we report the odds-ratios (OR), and for the Mann–Whitney U, we report the W test statistic. *p*-values were Bonferroni-corrected for multiple comparisons. All analyses were performed in R (R Core Team, 2015).

Genetic Analyses

Genotypes from the *M. × domestica* evaluated here were generated using GBS described in Gardner et al. (unpublished data, 2016). SNPs with a minor allele frequency (MAF) < 0.01 were excluded. Accessions with > 30% missing data were excluded and SNPs with < 20% missing data were retained. The resulting genotype matrix contained 8657 SNPs, 929 accessions, and 9.3% missing data. Missing genotypes were imputed using LinkImpute (Money et al., 2015). For imputation, we used values of 5 and 20 for parameters *l* and *k*, which resulted in an estimated genotype imputation accuracy of 92%.

To determine if two or more accessions were clonally related, we calculated identity-by-descent (IBD) using PLINK (Purcell et al., 2007; Purcell, 2009). When two or more accessions had IBD ($\hat{\pi}$) > 0.9, one accession from the clonal group was randomly chosen, and its genotype data were retained while the genotype data from the other clones were removed. The remaining dataset contained 840 accessions, including 689 with phenotype data.

For PCA, SNPs were pruned for LD using PLINK by considering a window of 10 SNPs, removing one SNP from a pair if LD > 0.5, then shifting the window by three SNPs and repeating the procedure (PLINK command: indep-pairwise 10 3 0.5). After removing SNPs with MAF < 0.05, 4395 SNPs and 672 accessions remained for PCA.

We calculated LD decay using PLINK, and used only SNPs with MAF > 0.05. The apple reference genome contains numerous large gaps of unknown sequence of varying length represented by long series of 'N's (Velasco et al., 2010). To avoid bias in our LD decay measures, we discarded LD estimates generated from SNP pairs separated by a gap > 10,000 'N's. There are 3590 gaps > 10 kb in the apple reference genome v1.0p (Genome Database for Rosaceae, GDR, available at www.rosaceae.org) (Jung et al., 2014).

A GWAS was performed using EMMAX (Kang et al., 2010). The *k* matrix was generated in EMMAX (command: emmax-kin-v-h-s-d 10) and we corrected for relatedness using the *k* matrix without any additional covariates. We used the GBrowse tool (GDR) (Jung et al., 2014) for *Malus × domestica* v1.0p to check for potential genes of interest near GWAS hits that passed the Bonferroni corrected threshold for significance (*p* < 0.05). We examined the distribution of phenotype data for the most significant GWAS SNPs and represented them using the tableplot package in R (Kwan and Friendly, 2012). We also tested which model of inheritance fit best based on

the single most significant SNP for overcolor intensity using SNPStats (Sole et al., 2006).

A significant GWAS result for firmness and harvest time was identified in a NAC protein and we aligned NAC proteins from various plant species using ClustalW (Larkin et al., 2007). A phylogenetic tree was built using MEGA6 with the Dayhoff model (Dayhoff et al., 1978) and neighbor joining method. We used a pairwise deletion option for dealing with gaps, and a consensus of 1000 bootstrap replicates (Tamura et al., 2013).

Genomic prediction was performed using the x.val function in the R package PopVar (Mohammadi et al., 2015). The rrBLUP model was selected and prediction accuracy was assessed using a fivefold (nFold = 5) cross-validation procedure, which masked 20% of the samples' phenotypes and then predicted them using a model generated from the remaining 80%. All other default parameters were used. Genomic prediction accuracy was calculated as the correlation between the predicted phenotypes and the observed values.

Results and Discussion

Historical Data Curation

We downloaded 121,950 phenotypic observations for the genus *Malus* from the USDA-GRIN database. These observations came from 105 different phenotypes measured in 4123 accessions spanning 15 yr. For the purposes of GWAS and genomic prediction, only accessions with shared segregating polymorphisms are useful. We therefore restricted the data to members of the domesticated apple, *M. × domestica*, which make up 32% (1339) of the accessions in the database. After filtering (Supplemental Fig. S1), the resulting dataset contained 24,778 measurements from 36 phenotypes across 689 different accessions, which represents approximately 20% of the data available for the genus *Malus* in the USDA-GRIN database. It is worth noting that only 36 of the initial 105 phenotypes (34%) were deemed useful for downstream analyses. Most phenotypes were not measured in enough accessions, or were not measured in an appropriate manner, to be useful for genetic mapping and genomic prediction.

Figure 1A shows the frequency of phenotypes according to amount of data available. Although promising on initial inspection, it does not account for the fact that data are often collected across multiple years. In only 26% of cases were there >100 data points for a given phenotype within a particular year. Sample sizes were often small and data collection was highly uneven across years, potentially due to available funding and access to resources (Fig. 1B). Of the 36 phenotypes included, seven were measured in a single year, while 24 were measured across 10 or more years (Fig. 1C). The bimodal distribution in Fig. 1C is the result of having a core set of phenotypes which were measured frequently, while the remaining phenotypes were measured once. Even when a phenotype was measured across multiple years, the same trees were often not

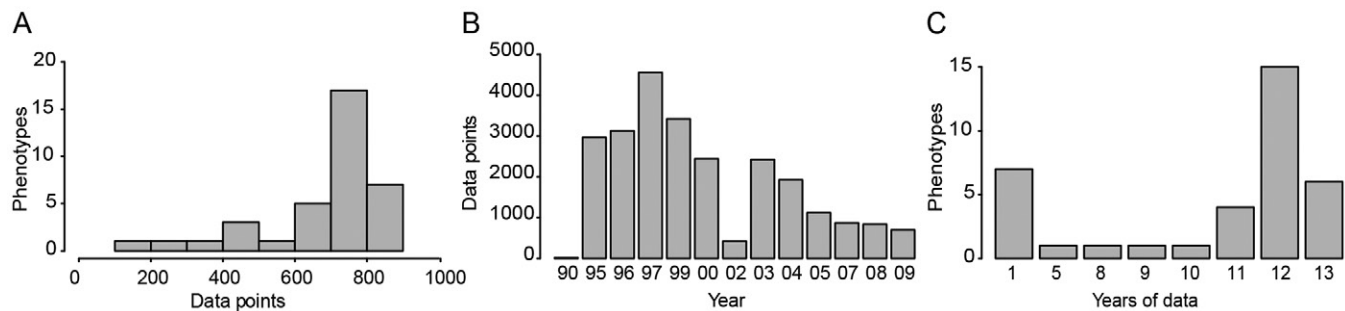


Fig. 1. Description of phenotype data available from USDA-GRIN database for accessions belonging to the domesticated *M. × domestica*. (A) Frequency of phenotypes according to number of data points available. (B) Number of data points by year. Years with no data available are not shown. (C) Number of years of data available for each phenotype; only values that apply to at least one phenotype are shown.

phenotyped each time, so the data are highly unbalanced and corrections for year effects could not be implemented.

Both inconsistent data collection across years and small sample sizes within years make exploring genotype-phenotype associations using historical data challenging. When an accession was measured across multiple years for a phenotype, we used the mean phenotype score across years in our analyses. Extensive data curation (Supplemental Table S1) was required to generate data sets that could be successfully linked to genotype data. Supplemental Fig. S1 provides a flow chart of how the data were handled from the initial database download to the final data set. A description of each phenotype and its phenotype scoring method are provided in Supplemental Table S1. The sample size and distribution of scores for the 36 phenotypes examined are shown in Supplemental Fig. S2.

Correlations among Phenotypes

Even without genotype data, assessing patterns within the phenotype data helps assess data reliability while potentially exposing novel relationships worthy of further inquiry. We therefore investigated correlations between all pairs of the 36 phenotypes remaining after data curation (Fig. 2). All *p*-values reported below are Bonferroni-corrected for multiple comparisons. The strongest correlations were between fruit length and width ($r = 0.850$, $p < 1 \times 10^{-15}$), fruit length and weight ($r = 0.517$, $p < 1 \times 10^{-15}$), and fruit weight and width ($r = 0.567$, $p < 1 \times 10^{-15}$). In these cases, an increase in one fruit size measurement is positively correlated with an increase in another, indicating that longer fruit are also heavier and wider.

Three different measurements evaluating the amount and location of russetting (rough, brown skin) were taken: intensity was measured as the percent of fruit surface covered in russet (0 to 100%), location of russetting indicated which area of the fruit was russeted (either one end, both ends, or entire fruit), and fruit russet type was a binary trait describing the russetting as either extremely fine or medium heavy to cracked. All measurements of fruit russetting, including intensity and location ($r = 0.370$, $p = 1.093 \times 10^{-15}$), intensity and type ($r = 0.378$, $p < 1 \times 10^{-15}$), and type and location ($r = 0.340$, $p = 4.725 \times 10^{-13}$), were

positively correlated with each other. Fruit overcolor was negatively correlated with all russetting measurements, indicating that green apples were more russeted than red apples (intensity: $r = -0.196$, $p = 3.792 \times 10^{-3}$; type: $r = -0.181$, $p = 0.050$; location: $r = -0.228$, $p = 3.402 \times 10^{-4}$). These correlations are all expected and provide confidence in the reliability of the phenotype data.

Thicker fruit stems were found to be shorter ($r = -0.207$, $p = 5.906 \times 10^{-5}$) and attached to heavier ($r = 0.164$, $p = 0.016$), longer ($r = 0.198$, $p = 2.150 \times 10^{-4}$), and wider fruits ($r = 0.249$, $p = 9.992 \times 10^{-8}$), potentially enabling larger, heavier apples to better remain attached to the tree.

Fruit uniformity in shape and size was scored as either uniform or variable by visually comparing 10 apples from the same cultivar. Fruit that were more uniform in shape were also more uniform in size ($r = 0.361$, $p < 1 \times 10^{-15}$). Uniformity facilitates the processing of commercial cultivars, thus the correlation between uniform size and shape may be due to selective breeding for this desirable trait (Hampson and Hemp, 2003).

In agreement with previous work (Jan et al., 2012), varieties harvested late in the season tended to be juicier ($r = 0.210$, $p = 0.007$) and have higher soluble solids ($r = 0.359$, $p = 2.142 \times 10^{-18}$). Harvest time was also negatively correlated with harvest uniformity ($r = 0.348$, $p = 3.090 \times 10^{-10}$), indicating that apples harvested earlier in the season required fewer visits to the tree during harvest. Staff being occupied by other activities early in the growing season may partially account for this observation. Accessions with heavier apples required more visits during harvest ($r = 0.382$, $p = 9.792 \times 10^{-13}$) and tended to have a coarser fruit texture ($r = 0.192$, $p = 4.981 \times 10^{-4}$).

Cultivars with more russetting locations ($r = -0.205$, $p = 1.183 \times 10^{-3}$) or a higher intensity of russetting ($r = -0.202$, $p = 4.887 \times 10^{-4}$) tended to have lower natural bloom, or wax, on the fruit at maturity. In our study, wax was scored as simply present or absent and did not distinguish further based on cuticle properties. However, wax impacts the cuticle of the fruit and, as a result, could play a role in the susceptibility to russetting, although this may be a complicated relationship (Khanal et al., 2013). In previous work, genes involved in cutin and wax

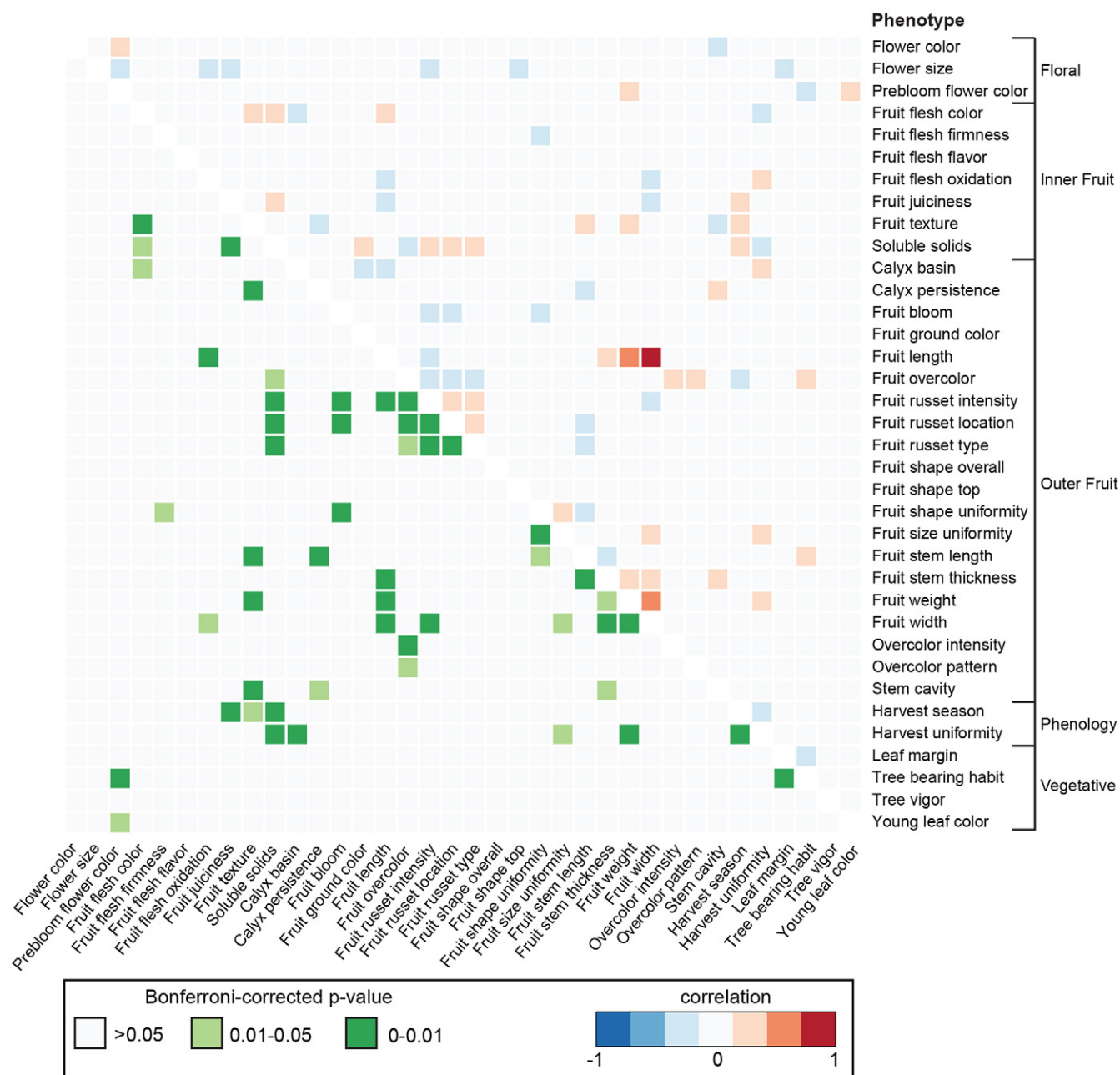


Fig. 2. Correlations among apple phenotypes. Values above the diagonal are colored to indicate the correlation results (r) and those below the diagonal indicate Bonferroni-corrected p -values.

synthesis were downregulated in russeted apple skin (Legay et al., 2015). Increased russetting type ($r = 0.260$, $p = 7.505 \times 10^{-7}$), location ($r = 0.276$, $p = 6.556 \times 10^{-8}$), and intensity ($r = 0.259$, $p = 1.246 \times 10^{-7}$), as well as later harvest season ($r = 0.359$, $p = 2.142 \times 10^{-18}$) were also correlated with higher soluble solids.

Differences between Apple Types

Cultivars were divided into several binary categories according to information from GRIN and/or an online search. Wherever possible, an accession was categorized as either from the Old World or the New World; as a primarily red or other (green/yellow) apple; as primarily used for cider or other (eating/cooking) purposes; and as

a late (October/November) or early (August/September) variety (Supplemental Table S2). All p -values reported below are Bonferroni-corrected for multiple comparisons.

Similar to the results in Fig. 2, we found that apples we scored as “late” tended to be juicier (OR = 3.714, $p = 0.006$) and have a higher soluble solids concentration ($W = 5590.5$, $p = 6.11 \times 10^{-7}$) than apples we scored as “early,” providing support for the accuracy of our scoring (Fig. 3). In agreement with previous work, firmness was also higher for later varieties (OR = 4.461, $p = 1.3 \times 10^{-6}$) (Watkins et al., 2000; Oraguzie et al., 2004; Nybom et al., 2012). We also found that red apples had a higher level of fruit bloom or wax on the fruit compared with green/yellow apples (OR = 0.473, $p = 0.007$).



Fig. 3. The relationship between apple categories and phenotypes. Each phenotype was divided into two groups according to various categories (harvest time, color, geography and use) and compared. *p*-values are Bonferroni-corrected.

New World apples were generally larger in size including length ($W = 30,705$, $p = 9.651 \times 10^{-10}$), width ($W = 31,220.5$, $p = 4.284 \times 10^{-7}$), and weight (OR = 1.958, $p = 0.010$), while Old World apples had a higher russet intensity ($W = 47,328.5$, $p = 1.937 \times 10^{-7}$) and were less likely to be red (OR = 2.626, $p = 1.791 \times 10^{-3}$). These observations suggest there may have been stronger selection in New World breeding programs for large red apples with less russetting. As expected based on previous work, apples primarily used for cider were smaller in fruit weight (OR = 5.791, $p = 1.966 \times 10^{-5}$), length ($W = 3034.5$, $p = 1.618 \times 10^{-10}$), and width ($W = 3158.5$, $p = 6.136 \times 10^{-9}$), compared with other (eating and cooking) apples (Miles and King, 2014). Cider apples also oxidized more than eating/cooking apples ($W = 10,860$, $p = 2.592 \times 10^{-6}$).

Population Structure

The genetic structure of the accessions from the USDA collection was investigated using PCA. Principal components (PCs) were calculated from the genome-wide SNP data. Accessions were plotted along the first two

PCs and labeled by harvest time and geography (Fig. 4A). The primary axis of genetic structure (PC1) in apple distinguishes early ripening from late ripening accessions (Fig. 4B) ($W = 5045.5$, $p = 2.441 \times 10^{-25}$) while Old World and New World apples differ along PC2 (Fig. 4C) ($W = 117,404.5$, $p = 1.317 \times 10^{-44}$) indicating that population structure within the domesticated apple is at least partially due to differences in origin and harvest time.

According to a PCA of genome-wide SNPs, cider and other (eating and cooking) varieties differ significantly along PC1 ($W = 8796$, $p = 7.082 \times 10^{-4}$) and PC2 ($W = 19,264$, $p = 8.048 \times 10^{-17}$; Supplemental Fig. S3). In contrast, two previous studies found weak genetic differentiation between cider and dessert cultivars (Cornille et al., 2012; Leforestier et al., 2015).

We also investigated the degree to which phenotypes were correlated with population structure. We determined the proportion of phenotypic variance (R^2) explained by the first 10 genetic PCs for each phenotype and values ranged from 0.07 to 28% (Supplemental Fig. S4). In agreement with Fig. 4, harvest season was most strongly correlated with population structure: the first

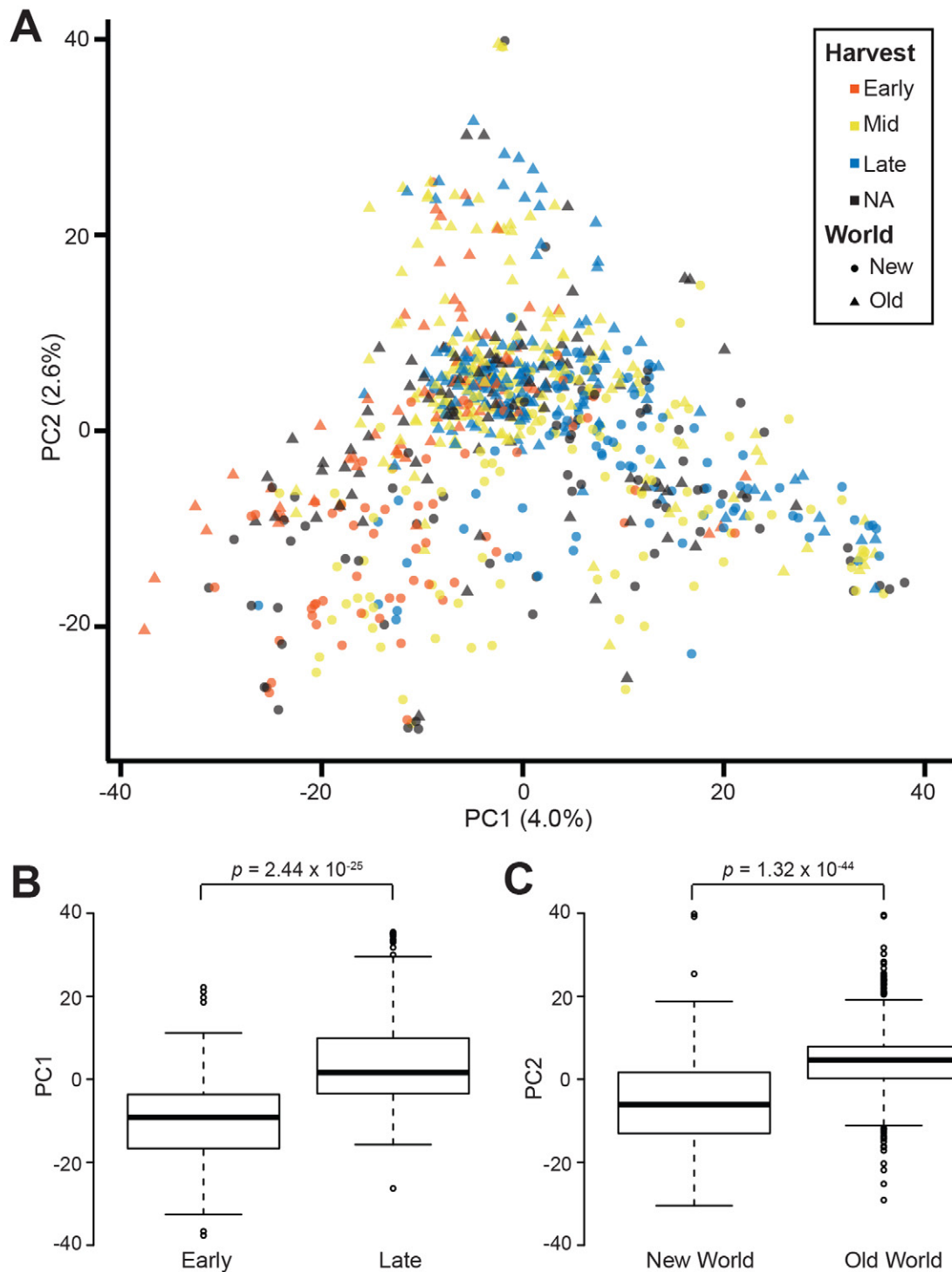


Fig. 4. Genetic relatedness based on harvest time and geographical origin. (A) Principal components (PC) analysis was performed using genome-wide SNP data. All samples with known geography information were retained and labeled based on geography with point shape (Old World vs. New World) as well as harvest time (early, midharvest, and late) with point color when possible. Unknown harvest times are marked as NA. The percentage of the variance explained by each PC is indicated in parentheses along each axis. (B) Boxplot of PC1 values for early vs. late harvest varieties of apples. (C) Boxplot of PC2 values for New and Old World varieties of apple. Results are reported from a Mann–Whitney U-test.

10 PCs explained 28% of the variance in harvest season, with PC1 explaining 16% of the variance. The harvest season of an accession is a proxy for the amount of time it requires for that accession to mature. Like many other phenological traits (e.g., Hall and Willis, 2006; Grillo

et al., 2013), this measure of ripening time has likely evolved in response to local climates. Thus, the harvest season of an apple may reflect, to some degree, its geographic ancestry. Geography is often a strong predictor of genetic relatedness, and models of isolation-by-distance

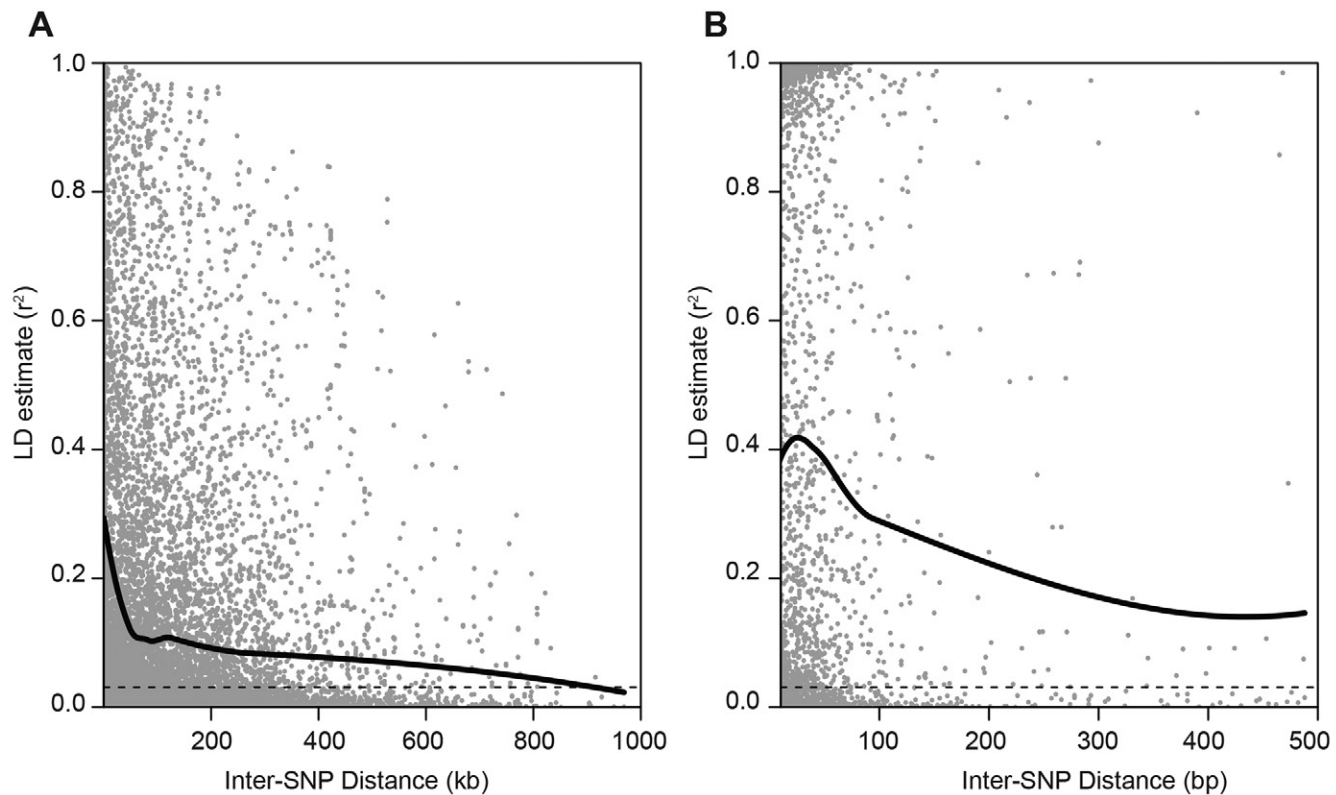


Fig. 5. Linkage disequilibrium (LD) decay curve in apple. (A) LD decay using comparisons of inter-SNP distances up to 1 Mb. (B) LD decay from comparisons of inter-SNP (single nucleotide polymorphism) distances of 10 to 500 bp. Smoothed fitted lines were calculated using the LOESS method. The horizontal dotted lines represent background LD: it is the upper 95% confidence interval from 10,000 LD measures generated from comparisons between SNP pairs from different chromosomes.

are commonly supported by population genomic data (e.g., Cao et al., 2011). Our observation of harvest season as the trait most strongly correlated with population structure in apples is consistent with a model of isolation-by-distance, with the notion that the time required for an accession to mature has been shaped by the geographic origin of its ancestors.

Phenotypic measures are often used as proxies of genetic relatedness: it is assumed that apples that are more closely related should also be more phenotypically similar. We investigated the extent to which a summary of all phenotype measurements captured the genetic relatedness among samples. To accomplish this, we calculated a phenotypic Euclidean distance using the `dist()` function in R among all pairs of cultivars. We used varying numbers of phenotypes and compared this pairwise distance matrix to a pairwise kinship matrix generated from the genome-wide SNP data using a Mantel test. In our study, phenotypic measurements captured at most 24% of the variance of the kinship matrix. Interestingly, increasing the number of phenotypes measured may decrease the ability to explain genetic relatedness among cultivars, potentially due to the inclusion of low-quality phenotypes (Supplemental Fig. S5). Thus, assessments of relatedness based on phenotype data alone are unlikely to accurately capture the genetic relatedness of germplasm collections such as this one. Inferences about relatedness in germplasm collections should make use of

genotype data and not be based on phenotype data alone whenever possible (Jansky et al., 2015).

LD and GWAS

The power of GWAS is dictated in part by the degree to which genotyped SNPs are in LD with causal alleles. We examined the extent of LD decay in the apple genome based on 4096 SNPs, which were 124 kb apart on average. Linkage disequilibrium in apple is generally low, even at close distances, and decays rapidly (Fig. 5A). We had 1900 SNP pairs which were <500 bp apart, and this allowed us to assess LD decay even at short distances (Fig. 5B). At distances of <100 bp, there is a bimodal distribution of LD, with many SNPs in high LD ($r^2 > 0.8$) and many in low LD ($r^2 < 0.2$). Very few SNPs have high LD when the inter-SNP distance is >100 bp.

The rapid LD decay we observe in apple is likely due to the lack of a true founder population resulting in high species diversity, as occurs in many perennial fruits such as the domesticated grape (*Vitis vinifera* L.) (Myles et al., 2010; Khan and Korban, 2012). Given that LD decays to $r^2 < 0.2$ at approximately 100 bp, we reason that, on average, a SNP is needed every 100 bp to perform well-powered GWAS in a diverse collection of domesticated apples. The apple has a genome size of ~750 Mb (Velasco et al., 2010), and we therefore estimate that millions of SNPs are needed for robust GWAS in diverse collections of apples. It also implies that GWAS in apple collections

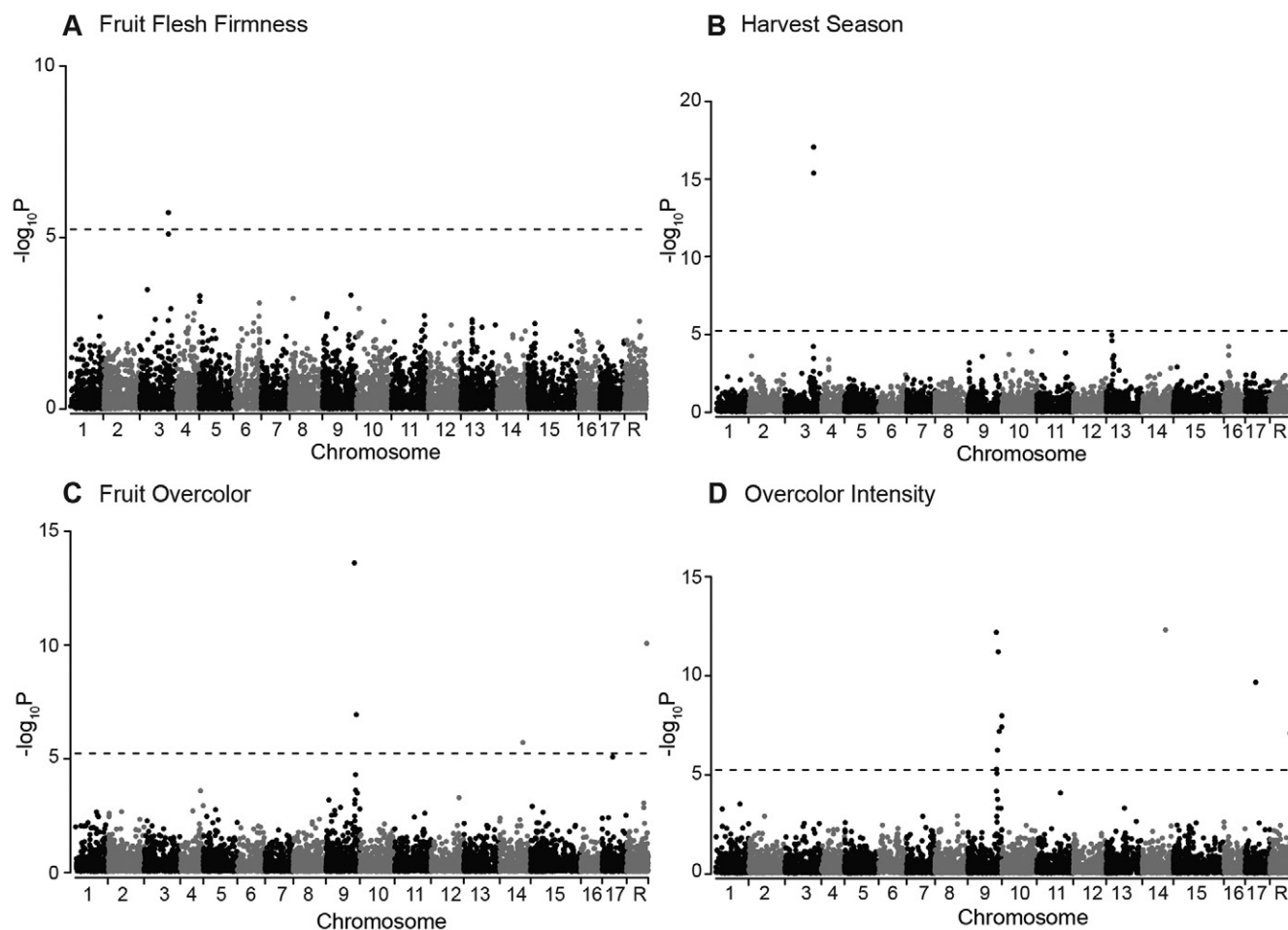


Fig. 6. Manhattan plot of genome-wide association study (GWAS) results for traits of interest, including (A) fruit flesh firmness, (B) harvest season, (C) fruit overcolor, and (D) overcolor intensity. p -values are log-transformed, and threshold for significance is Bonferroni-corrected and indicated by the horizontal dotted lines. Chromosome R indicates SNPs found on contigs that remain unanchored to the reference genome.

such as the one studied here will achieve extraordinarily high mapping resolution and it will likely be possible to localize the precise nucleotide positions of many causal variants without the need to perform additional fine mapping, which holds great promise for advancing both MAS and genome editing.

While rapid LD decay, relatively low SNP density and the use of historical phenotype data provided potential barriers to successful GWAS, we found several results of note (Fig. 6), including significant associations with fruit firmness (Fig. 6A), harvest time (Fig. 6B), and fruit color (Fig. 6C, 6D). Complete GWAS results for all 36 phenotypes examined can be found in Supplemental Fig. S6 and the alleles, p -value, MAF, and effect for all 31 SNPs significantly associated with a phenotype after Bonferroni correction are listed in Supplemental Table S3.

The apple is a climacteric fruit in which ethylene production during fruit ripening drives a loss of fruit firmness. Excessive softening is undesirable and leads to lower consumer acceptability, so there is a strong interest in breeding cultivars that retain their firmness during extended storage (Johnston et al., 2002). Cultivars harvested later in the season also tend to have a slower

softening rate, potentially due to having smaller cells and smaller intercellular spaces that may result in stronger tissue (Johnston et al., 2002; Nybom et al., 2012).

Alleles at two genes in ethylene's biosynthetic pathway, *Md-ACO1* on chromosome 10 and *Md-ACS1* on chromosome 15, have been repeatedly associated with fruit firmness (Oraguzie et al., 2004; Costa et al., 2005; Costa et al., 2010), and markers at these loci are used in marker-assisted breeding (Edge-Garza et al., 2010). We did not find significant associations with fruit firmness on chromosome 10 or 15, likely due to low SNP density surrounding the causal loci. While we were unable to determine the exact position of *Md-ACS1* in the reference genome, the SNPs closest to *Md-ACO1* were >100 kb away. Given the rapid extent of LD decay, it is not surprising we were unable to identify a significant peak for these two loci.

Fruit firmness is a complicated physiological process and in addition to *Md-ACO1* and *Md-ACS1*, other genes are also involved (Atkinson et al., 2012; Costa et al., 2014). Here, we report a single GWAS hit on chromosome 3 for firmness (Fig. 6A) that overlaps with the hit for harvest time (Fig. 6B; Supplemental Table S3). The overlap of GWAS hits for firmness and harvest time is

likely due to the physiological relationship between fruit maturity and firmness. Previous work, as well as our own (Fig. 3), have reported positive correlations between later harvest and firmer fruit (Oraguzie et al., 2004; Nybom et al., 2012).

Previous linkage mapping studies in biparental apple populations identified QTLs for both firmness and harvest date on chromosome 3 (Liebhard et al., 2003; Kenis et al., 2008). A recent GWAS also identified significant SNPs for fruit firmness on chromosome 3 (Kumar et al., 2013a). Our GWAS hit on chromosome 3 is consistent with these previously identified QTL for firmness and harvest date. Our high-resolution GWAS enabled a refinement of the position of this QTL on the distal end of chromosome 3, and we found that it falls within the coding region of NAC18.1 (GenBank ID: NM_001294055.1; chr3:31407982..31409374). NAC18.1 is a transcription factor that belongs to the NAC gene family, which is one of the largest families of plant-specific transcription factors (Olsen et al., 2005). NAC proteins are known to be involved in fruitlet abscission in apple (Botton et al., 2011) and ripening in peach [*Prunus persica* (L.) Batsch] (Pirona et al., 2013) and banana (*Musa acuminata* Colla) (Shan et al., 2012).

In our study, the most significant GWAS hit results in a nonsynonymous substitution from aspartic acid (D) to tyrosine (Y) at the fifth amino acid of NAC18.1, which we refer to as D5Y. Although D5Y does not appear to fall within a functional domain (de Castro et al., 2006), a D to Y amino acid substitution results in a score of -3 according to the BLOSUM62 matrix (Henikoff and Henikoff, 1992), suggesting it may be biologically significant. Most notably, D5Y lies within a motif that we refer to as the TDSS motif. Using phylogenetic analysis, we demonstrate that NAC proteins possessing a TDSS motif cluster together with a bootstrap value of 88. The NAC proteins showing the highest homology to NAC18.1 have the TDSS motif, and the D residue is conserved in 22 of the 25 proteins analyzed (Supplemental Fig. S7 and S8). We hypothesize that the possession of this motif indicates shared evolutionary function. For example, in tomato (*Solanum lycopersicum* L.) the NOR protein is required for ripening (Karlova et al., 2014). Likewise, in *Arabidopsis thaliana*, NAC2 is involved in ethylene-mediated senescence (Qiu et al., 2015). In kiwifruit (*Actinidia arguta* var. *arguta*), one member of the NAC protein family, NAC2, which encodes a variant of the TDSS motif (PDSS), activates the promoter of terpene synthase 1 (*TPS1*) more strongly than NAC1 and NAC3, which carry the conserved TDSS motif. *TPS1* expression results in aromatic terpene production in ripe fruit (Nieuwenhuizen et al., 2015). The D to Y substitution we describe here correlates with softer apples that ripen earlier. If D5Y is indeed causal, it may act as a gain of function by activating the expression of downstream proteins and accelerating ripening in apple. Further functional studies are required to reveal the possible function of D5Y.

In addition to the D5Y substitution identified, harvest season had two additional significant SNPs near the NAC18.1 coding region (chr3:31409376 and chr3:31409480, Supplemental Table S3), both in high LD ($r^2 > 0.94$) with D5Y. Ultimately, SNPs in and around NAC18.1 are potential markers for MAS, which would help determine harvest time and firmness in seedlings from new crosses with unknown maturity dates and ensure cultivars were grown in the appropriate climate to ripen before winter.

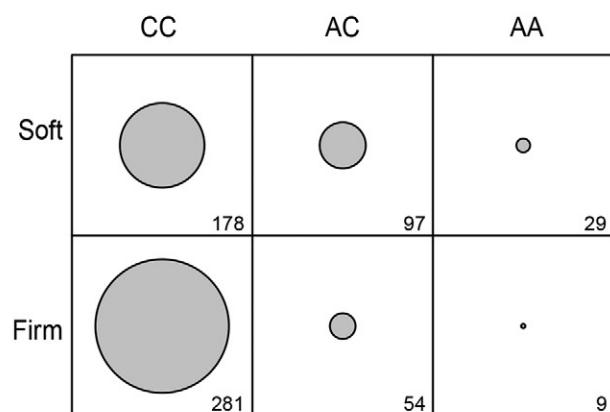
Identification of markers for fruit skin color in apple using GWAS may be useful for marker-assisted breeding by allowing for skin color selection during the juvenile phase, before fruit production (Zhang et al., 2014). Anthocyanins are responsible for the red coloration in apples and the transcription factor MdMYB1 regulates anthocyanin genes, forming the genetic basis for apple skin color (Talos et al., 2006; Ban et al., 2007). In the present study, we tested for associations with fruit overcolor (Fig. 6C), which is a binary assessment (i.e., red vs. green), and with overcolor intensity (Fig. 6D), which is a quantitative measurement of the percentage of overcolor (generally red) on a fruit (Supplemental Table S1). We confirmed the association between MYB1 and fruit color: the most convincing peaks for fruit overcolor and overcolor intensity occur near position 32.8 Mb on chromosome 9 where the MYB1 is located (Zhu et al., 2010; Gardner et al., 2014).

To investigate the ability of single GWAS hits (Fig. 6) to explain phenotypic variance, we examined the phenotype data for each of the three possible genotypes at each of the most significant GWAS SNPs (Fig. 7). We found that an accession with at least one minor allele (A) at chr3:31409362 was “soft” in 67% of cases (Fig. 7A). Accessions with this same allele had an “early” harvest time in 63% of cases (Fig. 7B). When using a binary assessment for fruit color, accessions with at least one minor allele (A) at position chr9:31448296 were “red” 87% of the time (Fig. 7C). Further studies are required to evaluate the utility of these markers for predicting phenotypes in other populations.

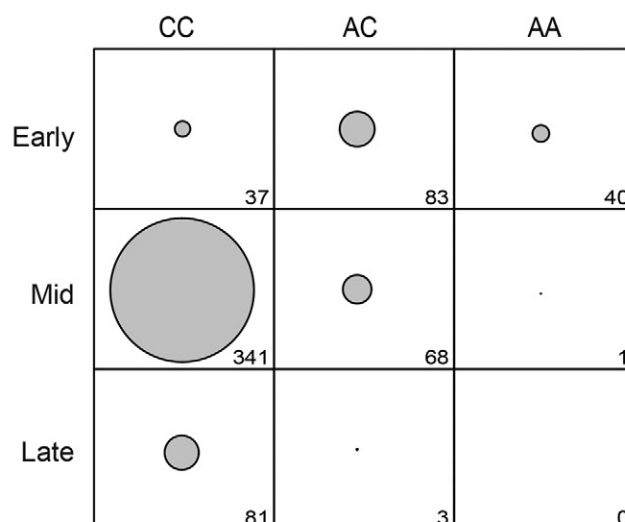
We also used quantitative measurements of fruit overcolor intensity to estimate the mode of inheritance using SNPStats (Fig. 7D) (Sole et al., 2006). Based on a single SNP, a codominant model fit best in which a single A allele resulted in a 22% increase in fruit overcolor intensity, while two A alleles resulted in a 28% increase compared with the mean overcolor value of TT (Akaike Information Criterion [AIC] = 5684). This codominant model fits only slightly better than a dominant model (AIC = 5692) in which a single A allele results in the full phenotypic effect. Based on our dataset, these SNPs all have potential for MAS and the ability to help select for desirable characteristics such as firmness and color using a single SNP.

Significant GWAS SNPs that are relatively isolated and not near any other significant SNPs may still be true positives given that we observed such rapid LD decay. However, some of the genotype-phenotype associations may be either false positives or simply mismapped SNPs

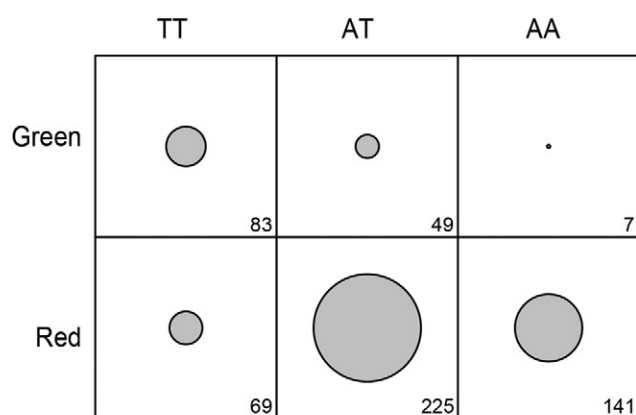
A Fruit Flesh Firmness chr3:31409362



B Harvest Season chr3:31409362



C Fruit Overcolor chr9:31448296



D Overcolor Intensity chr9:31448296

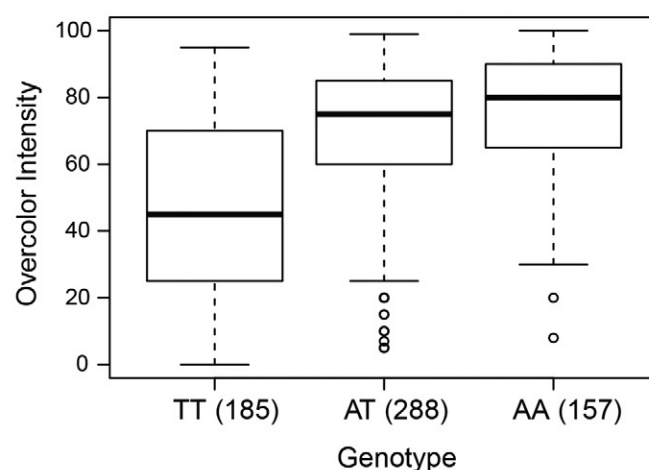


Fig. 7. Distribution of phenotype scores stratified by genotype at the most significant genome-wide association study single nucleotide polymorphisms for (A) firmness, (B) harvest season, (C) fruit overcolor, and (D) overcolor intensity. The sequence for each potential genotype is indicated. The number of observations within a particular genotype or phenotype category is listed. Circled areas are proportional to the number of observations.

in the reference genome. For example, LD between the top hit for color on chromosome 9 and the lone hit on chromosome 17 is as strong as the LD between the top chromosome 9 hit and many physically adjacent SNPs that fall within the peak overlapping the known causal locus. Therefore, the chromosome 17 SNP may have been assigned an incorrect physical position in the reference genome sequence used in this study. In two previous linkage mapping studies, 13.7 and 18.3% of SNPs were assigned to linkage groups that conflicted with the predicted chromosomal locations according to the reference genome (Antanaviciute et al., 2012; Gardner et al., 2014). Caution is therefore warranted in drawing strong conclusions about the number of QTL or the genetic architecture of a trait when dealing with incomplete or poor quality reference genome sequences. The associations

we report in Supplemental Fig. S6 should be interpreted while taking this into consideration.

Phenotype Prediction Accuracy Using Historical Data in Apple

Although the rapid LD decay we observe here presents challenges for GWAS in apple using relatively low-density genotyping platforms such as GBS, we still observed suggestive hits, including several that overlap with known QTL. However, the strict thresholds of GWAS only enable the detection of loci of relatively large effect. Most traits of interest to breeders are arguably highly polygenic, and thus controlled by numerous small-effect loci. As genome-wide marker data become available for an increasing number of organisms, it is becoming common to evaluate the extent to which phenotypes can be

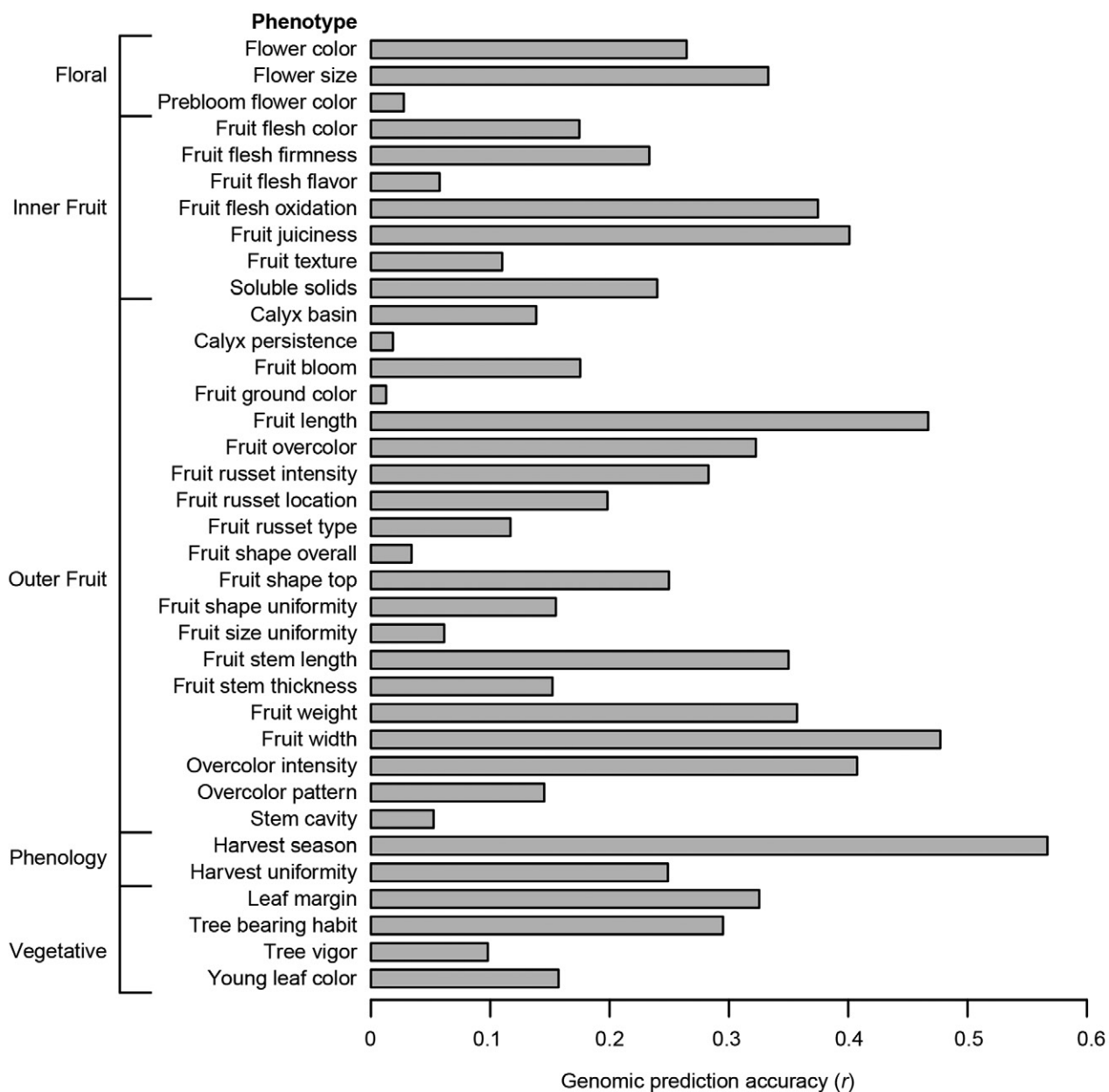


Fig. 8. Genomic prediction accuracy: r values represent the correlation between observed and predicted phenotype scores from genomic prediction using a five-fold cross-validation procedure.

predicted using genomic prediction methods (Heffner et al., 2011). Genomic prediction is a particularly useful tool for researchers interested in predicting complex, polygenic phenotypes, as it uses all markers simultaneously to predict phenotypes without identifying QTL (Gibson, 2010; Endelman, 2011).

To investigate the degree to which we can predict the phenotypes explored in this study, we performed genomic prediction analysis and determined that the highest prediction accuracy (0.57) was found for harvest season followed by fruit width (0.48) and length (0.47; Fig. 8). Similar prediction accuracies ranging from 0.31 for grain yield to 0.63 for flowering time have been found in rice (Spindel et al., 2015). Although many phenotypes are not well-predicted using genome-wide SNP data, or were collected in a way that does not enable genetic

mapping, this was not always the case. Several phenotypes appeared to be heritable, predictable, and even controlled by loci that were detected using GWAS. For example, color had fairly high prediction accuracies of 0.41 for overcolor intensity and 0.32 for fruit overcolor.

Phenotype prediction accuracy was highly correlated with the proportion of phenotypic variance explained by genetic PCs 1 to 10 ($r = 0.898$, $p = 1.141 \times 10^{-13}$) (Supplemental Fig. S9). While genetic PCs capture the principal axes of population genetic structure, phenotype prediction relies on a genetic relationship matrix among all samples derived from all SNPs. The observed positive relationship between these two methods of explaining phenotypic variation is expected, given that both methods capture genetic relatedness among accessions in different manners.

Conclusions

There are many difficulties associated with the use of historical phenotype data for association mapping and genomic prediction, including small sample sizes, use of subjective measurements, and inconsistent data collection across years. However, by recoding phenotype data and combining across years, we were able to observe relationships between phenotypes as well as provide evidence of several GWAS hits including color, fruit firmness, and harvest time. In particular, we report a novel nonsynonymous SNP in the transcription factor NAC18.1, which is correlated with softer apples that ripen earlier and warrants further functional investigation. The continued advent of high-throughput phenotyping technologies and improvement in phenotype measurement collection will increase our ability to understand the genome–phenome relationship in apple. Due to rapid LD decay, ultimately, whole genome sequencing may be required to enable well-powered association mapping in diverse collections like the one studied here. However, using sufficiently dense genotype data, we expect loci to be detected at nearly nucleotide resolution, which will allow for more widespread adoption of MAS.

Supplemental Information Available

Supplemental Fig. S1. Flowchart of processing for phenotype data.

Supplemental Fig. S2. Distributions of phenotype scores.

Supplemental Fig. S3. Cider was compared to Other (Eating and Cooking) varieties using PCA of genome-wide SNPs.

Supplemental Fig. S4. Proportion of variance explained for different phenotypes using PCs 1 to 10.

Supplemental Fig. S5. Comparison of distance among samples calculated from phenotype and genotype data.

Supplemental Fig. S6. GWAS results for all 36 phenotypes examined.

Supplemental Fig. S7. Multiple species alignment of NAC proteins.

Supplemental Fig. S8. Phylogenetic analysis of NAC protein family members.

Supplemental Fig. S9. Correlation between prediction accuracy and proportion of phenotypic variance explained by the first 10 genetic PCs.

Supplemental Table 1. Original and new scores for USDA-GRIN apple phenotype data, as well as USDA-GRIN trait descriptions.

Supplemental Table 2. Metadata including color, use, and harvest season for cultivars sequenced.

Supplemental Table 3. Position information for significant GWAS hits as well as *p*-value and MAF.

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References

- Antanaviciute, L., F. Fernández-Fernández, J. Jansen, E. Banchi, K.M. Evans, R. Viola, R. Velasco, J.M. Dunwell, M. Troggo, and D.J. Sargent. 2012. Development of a dense SNP-based linkage map of an apple rootstock progeny using the *Malus* Infinium whole genome genotyping array. *BMC Genomics* 13:203. doi:10.1186/1471-2164-13-203
- Atkinson, R.G., P.W. Sutherland, S.L. Johnston, K. Gunaseelan, I.C. Hallett, D. Mitra, D.A. Brummell, R. Schröder, J.W. Johnston, and R.J. Schaffer. 2012. Down-regulation of *POLYGALACTURONASE1* alters firmness, tensile strength and water loss in apple (*Malus × domestica*) fruit. *BMC Plant Biol.* 12:129. doi:10.1186/1471-2229-12-129
- Baldwin, S.J., K.G. Dodds, B. Auvray, R.A. Genet, R.C. Macknight, and J.M.E. Jacobs. 2011. Association mapping of cold-induced sweetening in potato using historical phenotypic data. *Ann. Appl. Biol.* 158:248–256. doi:10.1111/j.1744-7348.2011.00459.x
- Ban, Y., C. Honda, Y. Hatsuyama, M. Igarashi, H. Bessho, and T. Moriguchi. 2007. Isolation and functional analysis of a MYB transcription factor gene that is a key regulator for the development of red coloration in apple skin. *Plant Cell Physiol.* 48:958–970. doi:10.1093/pcp/pcm066
- Benfey, P.N., and T. Mitchell-Olds. 2008. From genotype to phenotype: Systems biology meets natural variation. *Science* 320:495–497. doi:10.1126/science.1153716
- Botton, A., G. Eccher, C. Forcato, A. Ferrarini, M. Begheldo, M. Zermiani, S. Moscatello, A. Battistelli, R. Velasco, B. Ruperti, and A. Ramina. 2011. Signaling pathways mediating the induction of apple fruitlet abscission. *Plant Physiol.* 155:185–208. doi:10.1104/pp.110.165779
- Brown, S.K., and K.E. Maloney. 2003. Genetic improvement of apple: Breeding, markers, mapping and biotechnology. In: D.C. Ferree and I.J. Warrington, editors. *Apples: Botany, production and uses*. CABI Publ., Oxfordshire, UK. p. 31–60.
- Burleigh, J. G., K. Alphonse, A. J. Alverson, H. M. Bik, C. Blank et al. 2013. Next-generation phenomics for the Tree of Life. *PLOS Currents Tree of Life*. 2013 June 26. Ed. 1. doi:10.1371/currents.tol.085c713acafc8711b2ff7010a4b03733
- Cao, J., K. Schneeberger, S. Ossowski, T. Gunther, S. Bender, J. Fitt, D. Koenig, C. Lanz, O. Stegle, C. Lippert, X. Wang, F. Ott, J. Muller, C. Alonso-Blanco, K. Borgwardt, K.J. Schmid, and D. Weigel. 2011. Whole-genome sequencing of multiple *Arabidopsis thaliana* populations. *Nat. Genet.* 43:956–963. doi:10.1038/ng.911
- de Castro, E., C.J. Sigris, A. Gattiker, V. Bulliard, P.S. Langendijk-Genevaux, E. Gasteiger, A. Bairoch, and N. Hulo. 2006. ScanProsite: Detection of PROSITE signature matches and ProRule-associated functional and structural residues in proteins. *Nucleic Acids Res.* 34:W362–W365. doi:10.1093/nar/gkl124
- Cobb, J.N., G. Declerck, A. Greenberg, R. Clark, and S. McCouch. 2013. Next-generation phenotyping: Requirements and strategies for enhancing our understanding of genotype-phenotype relationships and its relevance to crop improvement. *Theor. Appl. Genet.* 126:867–887. doi:10.1007/s00122-013-2066-0

- Cornille, A., P. Gladieux, M.J. Smulders, I. Roldan-Ruiz, F. Laurens, B. Le Cam, A. Nersesyan, J. Clavel, M. Olonova, L. Feugey, I. Gabrielyan, X.G. Zhang, M.I. Tenaillon, and T. Giraud. 2012. New insight into the history of domesticated apple: Secondary contribution of the European wild apple to the genome of cultivated varieties. *PLoS Genet.* 8:E1002703. doi:10.1371/journal.pgen.1002703
- Costa, F., L. Cappellin, B. Farneti, A. Tadiello, A. Romano, C. Soukoulis, S. Sansavini, R. Velasco, and F. Biasioli. 2014. Advances in QTL mapping for ethylene production in apple (*Malus × domestica* Borkh.). *Postharvest Biol. Technol.* 87:126–132. doi:10.1016/j.postharvbio.2013.08.013
- Costa, F., C.P. Peace, S. Stella, S. Serra, S. Musacchi, M. Bazzani, S. Sansavini, and W.E. Van de Weg. 2010. QTL dynamics for fruit firmness and softening around an ethylene-dependent polygalacturonase gene in apple (*Malus × domestica* Borkh.). *J. Exp. Bot.* 61:3029–3039. doi:10.1093/jxb/erq130
- Costa, F., S. Stella, W.E. Van de Weg, W. Guerra, M. Cecchinell, J. Dallavia, B. Koller, and S. Sansavini. 2005. Role of the genes *Md-ACO1* and *Md-ACS1* in ethylene production and shelf life of apple (*Malus domestica* Borkh.). *Euphytica* 141:181–190. doi:10.1007/s10681-005-6805-4
- Dayhoff, M.O., R.M. Schwartz, and B.C. Orcutt. 1978. A model of evolutionary change in proteins. In: M.O. Dayhoff, *Atlas of protein sequence and structure*. Vol. 5. National Biomedical Research Foundation, Washington, DC. p. 345–352.
- Deans, A.R., S.E. Lewis, E. Huala, S.S. Anzaldo, M. Ashburner et al. 2015. Finding our way through phenotypes. *PLoS Biol.* 13:E1002033. doi:10.1371/journal.pbio.1002033
- Edge-Garza, D.A., C.P. Peace, and Y. Zhu. 2010. Enabling marker-assisted seedling selection in the Washington apple breeding program. *Acta Hortic.* 859:369–373. doi:10.17660/ActaHortic.2010.859.44
- Elshire, R.J., J.C. Glaubitz, Q. Sun, J.A. Poland, K. Kawamoto, E.S. Buckler, and S.E. Mitchell. 2011. A robust, simple genotyping-by-sequencing (GBS) approach for high diversity species. *PLoS ONE* 6:E19379. doi:10.1371/journal.pone.0019379
- Endelman, J.B. 2011. Ridge regression and other kernels for genomic selection with R package rrBLUP. *Plant Gen.* 4:250. doi:10.3835/plantgenome2011.08.0024
- Fazio, G., Y. Wan, D. Kvikly, L. Romero, R. Adams, D. Strickland, and T. Robinson. 2014. Dw2, a new dwarfing locus in apple rootstocks and its relationship to induction of early bearing in apple scions. *J. Am. Soc. Hortic. Sci.* 139:87–98.
- Food and Agriculture Organization of the United Nations. 2015. FAOSTAT. Available at http://faostat3.fao.org/browse/rankings/commodities_by_regions/E (accessed 19 Oct. 2015; verified 15 Mar. 2016).
- Furbank, R.T., and M. Tester. 2011. Phenomics—Technologies to relieve the phenotyping bottleneck. *Trends Plant Sci.* 16:635–644. doi:10.1016/j.tplants.2011.09.005
- Gardner, K.M., P. Brown, T.F. Cooke, S. Cann, F. Costa, C. Bustamante, R. Velasco, M. Troggo, and S. Myles. 2014. Fast and cost-effective genetic mapping in apple using next-generation sequencing. *G3: Genes, Genomes, Genet.* 4:1681–1687.
- Gibson, G. 2010. Hints of hidden heritability in GWAS. *Nat. Genet.* 42:558–560. doi:10.1038/ng0710-558
- Grillo, M.A., C. Li, M. Hammond, L. Wang, and D.W. Schemske. 2013. Genetic architecture of flowering time differentiation between locally adapted populations of *Arabidopsis thaliana*. *New Phytol.* 197:1321–1331. doi:10.1111/nph.12109
- Hall, M.C., and J.H. Willis. 2006. Divergent selection on flowering time contributes to local adaptation in *Mimulus guttatus* populations. *Evolution* 60:2466–2477. doi:10.1554/05-688.1
- Hampson, C.R., and H. Hemp. 2003. Characteristics of Important Commercial Apple Cultivars. In: D. C. Ferree and I. J. Warrington, editors. *Apples: Botany, production and uses*. CABI Publ., Oxfordshire, UK. p. 61–89.
- Heffner, E.L., J.-L. Jannink, and M.E. Sorrells. 2011. Genomic selection accuracy using multifamily prediction models in a wheat breeding program. *Plant Gen.* 4:65–75. doi:10.3835/plantgenome.2010.12.0029
- Henikoff, S., and J.G. Henikoff. 1992. Amino acid substitution matrices from protein blocks. *Proc. Natl. Acad. Sci. USA* 89:10,915–10,919. doi:10.1073/pnas.89.22.10915
- Houle, D., D.R. Govindaraju, and S. Omholt. 2010. Phenomics: The next challenge. *Nat. Rev. Genet.* 11:855–866. doi:10.1038/nrg2897
- Jan, I., A. Rab, and M. Sajid. 2012. Storage performance of apple cultivars harvest at different stages of maturity. *J. Anim. Plant Sci.* 22:438–447.
- Jansky, S.H., J. Dawson, and D.M. Spooner. 2015. How do we address the disconnect between genetic and morphological diversity in germ-plasm collections? *Am. J. Bot.* 102:1213–1215. doi:10.3732/ajb.1500203
- Johnston, J.W., E.W. Hewett, and M.L.A.T.M. Hertog. 2002. Postharvest softening of apple (*Malus domestica*) fruit: A review. *N. Z. J. Crop Hortic. Sci.* 30:145–160. doi:10.1080/01140671.2002.9514210
- Jung, S., S.P. Ficklin, T. Lee, C.H. Cheng, A. Blenda, P. Zheng, J. Yu, A. Bombarely, I. Cho, S. Ru, K. Evans, C. Peace, A.G. Abbott, L.A. Mueller, M.A. Olmstead, and D. Main. 2014. The genome database for Rosaceae (GDR): Year 10 update. *Nucleic Acids Res.* 42:D1237–D1244. doi:10.1093/nar/gkt1012
- Kang, H. M., J.H. Sul, S.K. Service, N.A. Zaitlen, S.-Y. Kong, N.B. Freimer, C. Sabatti, and E. Eskin. 2010. Variance component model to account for sample structure in genome-wide association studies. *Nat. Genet.* 42:348–354. doi:10.1038/ng.548
- Karlova, R., N. Chapman, K. David, G.C. Angenent, G.B. Seymour, and R.A. de Maagd. 2014. Transcriptional control of fleshy fruit development and ripening. *J. Exp. Bot.* 65:4527–4541. doi:10.1093/jxb/eru316
- Kenis, K., J. Keulemans, and M.W. Davey. 2008. Identification and stability of QTLs for fruit quality traits in apple. *Tree Genet. Genomes* 4:647–661. doi:10.1007/s11295-008-0140-6
- Khan, M.A., and S.S. Korban. 2012. Association mapping in forest trees and fruit crops. *J. Exp. Bot.* 4045–4060. doi:10.1093/jxb/ers105
- Khanal, B.P., R. Shrestha, L. Hückstädt, and M. Knoche. 2013. Russetting in apple seems unrelated to the mechanical properties of the cuticle at maturity. *HortScience* 48:1135–1138.
- Kumar, S., M.C.A.M. Bink, R.K. Volz, V.G.M. Bus, and D. Chagné. 2012. Towards genomic selection in apple (*Malus × domestica* Borkh.) breeding programmes: Prospects, challenges and strategies. *Tree Genet. Genomes* 8:1–14. doi:10.1007/s11295-011-0425-z
- Kumar, S., D. Garrick, M. Bink, C. Whitworth, D. Chagne, and R. Volz. 2013a. Novel genomic approaches unravel genetic architecture of complex traits in apple. *BMC Genomics* 14:393. doi:10.1186/1471-2164-14-393
- Kumar, S., R.K. Volz, D. Chagne, and S. Gardiner. 2013b. Breeding for apple (*Malus × domestica* Borkh.) fruit quality traits in the genomics era. In: R. Tuberosa et al., editors, *Genomics of plant genetic resources: Volume 2. Crop productivity, food security and nutritional quality*. Springer, New York. p. 387–416.
- Kwan, E., and M. Friendly. 2012. Tableplot: Represents tables as semi-graphic displays, R package v.0.3-5. R Foundation for Statistical Computing, Vienna, Austria.
- Larkin, M.A., G. Blackshields, N.P. Brown, R. Chenna, P.A. McGettigan, H. McWilliam, F. Valentin, I.M. Wallace, A. Wilm, R. Lopez, J.D. Thompson, T.J. Gibson, and D.G. Higgins. 2007. Clustal W and Clustal X version 2.0. *Bioinformatics* 23:2947–2948. doi:10.1093/bioinformatics/btm404
- Leforestier, D., E. Ravon, H. Muranty, A. Cornille, C. Lemaire et al. 2015. Genomic basis of the differences between cider and dessert apple varieties. *Evol. Appl.* 8(7):650–661. doi:10.1111/eva.12270
- Legay, S., G. Guerriero, A. Deleruelle, M. Lateur, D. Evers, C.M. Andre, and J.F. Hausman. 2015. Apple russetting as seen through the RNA-seq lens: Strong alterations in the exocarp cell wall. *Plant Mol. Biol.* 88:21–40. doi:10.1007/s11103-015-0303-4
- Liebhart, R., M. Kellerhals, W. Pfammatter, M. Jertmini, and C. Gessler. 2003. Mapping quantitative physiological traits in apple (*Malus × domestica* Borkh.). *Plant Mol. Biol.* 52(3):511–526. doi:10.1023/A:1024886500979
- Lusser, M., C. Parisi, D. Plan, and E. Rodriguez-Cerezo. 2012. Deployment of new biotechnologies in plant breeding. *Nat. Biotechnol.* 30:231–239. doi:10.1038/nbt.2142
- Matthies, I.E., M. Malosetti, M.S. Roder, and F. van Eeuwijk. 2014. Genome-wide association mapping for kernel and malting quality traits using historical European barley records. *PLoS ONE* 9:E110046. doi:10.1371/journal.pone.0110046

- McClure, K.A., J. Sawler, K.M. Gardner, D. Money, and S. Myles. 2014. Genomics: A potential panacea for the perennial problem. *Am. J. Bot.* 101:1780–1790. doi:10.3732/ajb.1400143
- McCouch, S., G.J. Baute, J. Bradeen, P. Bramel, P.K. Bretting et al. 2013. Agriculture: Feeding the future. *Nature* 499:23–24. doi:10.1038/499023a
- McCouch, S.R., K.L. McNally, W. Wang, and R. Sackville Hamilton. 2012. Genomics of gene banks: A case study in rice. *Am. J. Bot.* 99:407–423. doi:10.3732/ajb.1100385
- Meneses, C., and A. Orellana. 2013. Using genomics to improve fruit quality. *Biol. Res.* 46:347–352. doi:10.4067/S0716-97602013000400006
- Miles, C.A., and J. King. 2014. Yield, labor, and fruit and juice quality characteristics of machine and hand-harvested 'Brown Snout' specialty cider apple. *HortTechnology* 24:519–526.
- Mohammadi, M., T. Tiede, and K.P. Smith. 2015. PopVar: A genome-wide procedure for predicting genetic variance and correlated response in biparental breeding populations. *Crop Sci.* 55:2068. doi:10.2135/cropsci2015.01.0030
- Money, D., K. Gardner, Z. Migicovsky, H. Schwaninger, G.Y. Zhong, and S. Myles. 2015. LinkImpute: Fast and accurate genotype imputation for nonmodel organisms. *G3: Genes, Genomes, Genet.* 5(11):2383–2390. doi:10.1534/g3.115.021667
- Morrell, P.L., E.S. Buckler, and J. Ross-Ibarra. 2011. Crop genomics: Advances and applications. *Nat. Rev. Genet.* 13:85–96.
- Myles, S. 2013. Improving fruit and wine: What does genomics have to offer? *Trends Genet.* 29:190–196. doi:10.1016/j.tig.2013.01.006
- Myles, S., J.-M. Chia, B. Hurwitz, C. Simon, G.Y. Zhong, E. Buckler, and D. Ware. 2010. Rapid Genomic Characterization of the genus *Vitis*. *PLoS ONE* 5:E8219. doi:10.1371/journal.pone.0008219
- Myles, S., J. Peiffer, P.J. Brown, E.S. Ersoz, Z. Zhang, D.E. Costich, and E.S. Buckler. 2009. Association mapping: Critical considerations shift from genotyping to experimental design. *Plant Cell* 21:2194–2202. doi:10.1105/tpc.109.068437
- Nieuwenhuizen, N.J., X. Chen, M.Y. Wang, A.J. Matich, R.L. Perez, A.C. Allan, S.A. Green, and R.G. Atkinson. 2015. Natural variation in monoterpene synthesis in kiwifruit: Transcriptional regulation of terpene synthases by NAC and ETHYLENE-INSENSITIVE3-like transcription factors. *Plant Physiol.* 167:1243–1258. doi:10.1104/pp.114.254367
- Nybom, H., M. Ahmadi-Afzadi, J. Sehic, and M. Hertog. 2012. DNA marker-assisted evaluation of fruit firmness at harvest and post-harvest fruit softening in a diverse apple germplasm. *Tree Genet. Genomes* 9:279–290. doi:10.1007/s11295-012-0554-z
- Olsen, A.N., H.A. Ernst, L.L. Leggio, and K. Skriver. 2005. NAC transcription factors: Structurally distinct, functionally diverse. *Trends Plant Sci.* 10:79–87. doi:10.1016/j.tplants.2004.12.010
- Oraguzie, N.C., H. Iwanami, J. Soejima, T. Harada, and A. Hall. 2004. Inheritance of the *Md-ACS1* gene and its relationship to fruit softening in apple (*Malus × domestica* Borkh.). *Theor. Appl. Genet.* 108:1526–1533. doi:10.1007/s00122-003-1574-8
- Peace, C., and J. Norelli. 2009. Genomics approaches to crop improvement in the Rosaceae. In: K.M. Folta and S.E. Gardiner, editors, *Genetics and genomics of Rosaceae*. Springer, New York. p. 19–53.
- Peil, A., F. Dunemann, K. Richter, M. Hoefer, I. Király, H. Flachowsky, and M.-V. Hanke. 2008. Resistance breeding in apple at Dresden-Pillnitz. In: 13th Int. Conf. on Cultivation Technique and Phytopathological Problems in Organic Fruit-Growing, Weinsberg, Germany. 18–20 Feb. 2008. p. 220–225.
- Pirona, R., I. Eduardo, I. Pacheco, D.S.L. Cassia, M. Miculan, I. Verde, S. Tartarini, L. Dondini, G. Pea, and D. Bassi. 2013. Fine mapping and identification of a candidate gene for a major locus controlling maturity date in peach. *BMC Plant Biol.* 13:166. doi:10.1186/1471-2229-13-166
- Purcell, S. 2009. PLINK v. 1.07. Whole genome association analysis toolset. Available at <http://pngu.mgh.harvard.edu/~purcell/plink/> (verified 15 Mar. 2016).
- Purcell, S., B. Neale, K. Todd-Brown, L. Thomas, M.A.R. Ferreira, D. Bender, J. Maller, P. Sklar, P.I.W. de Bakker, M.J. Daly, and P.C. Sham. 2007. PLINK: A toolset for whole-genome association and population-based linkage analysis. *Am. J. Hum. Genet.* 81:559–575. doi:10.1086/519795
- Qiu, K., Z. Li, Z. Yang, J. Chen, S. Wu, X. Zhu, S. Gao, J. Gao, G. Ren, B. Kuai, and X. Zhou. 2015. EIN3 and ORE1 accelerate degreening during ethylene-mediated leaf senescence by directly activating chlorophyll catabolic genes in *Arabidopsis*. *PLoS Genet.* 11:E1005399. doi:10.1371/journal.pgen.1005399
- R Core Team. 2015. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Ru, S., D. Main, K. Evans, and C. Peace. 2015. Current applications, challenges, and perspectives of marker-assisted seedling selection in Rosaceae tree fruit breeding. *Tree Genet. Genomes* 11:8. doi:10.1007/s11295-015-0834-5
- Shan, W., K. Jian-fei, L. Chen, H. Xie, P. Huan-huan, X. Yun-yi, L. Xue-ping, C. Wei-xin, H. Quan-guang, C. Jian-ye, and L. Wang-jin. 2012. Molecular characterization of banana NAC transcription factors and their interactions with ethylene signalling component EIL during fruit ripening. *J. Exp. Bot.* 63:5171–5187. doi:10.1093/jxb/ers178
- Sole, X., E. Guino, J. Valls, R. Iniesta, and V. Moreno. 2006. SNPStats: A web tool for the analysis of association studies. *Bioinformatics* 22:1928–1929. doi:10.1093/bioinformatics/btl268
- Spindel, J., H. Begum, D. Akdemir, P. Virk, B. Collard, E. Redona, G. Atlin, J.L. Jannink, and S.R. McCouch. 2015. Genomic selection and association mapping in rice (*Oryza sativa*): Effect of trait genetic architecture, training population composition, marker number and statistical model on accuracy of rice genomic selection in elite, tropical rice breeding lines. *PLoS Genet.* 11:E1004982. doi:10.1371/journal.pgen.1004982
- Takos, A.M., F.W. Jaffe, S.R. Jacob, J. Bogs, S.P. Robinson, and A.R. Walker. 2006. Light-induced expression of a *MYB* gene regulates anthocyanin biosynthesis in red apples. *Plant Physiol.* 142:1216–1232. doi:10.1104/pp.106.088104
- Tamura, K., G. Stecher, D. Peterson, A. Filipski, and S. Kumar. 2013. MEGA6: Molecular Evolutionary Genetics Analysis v.6.0. *Mol. Biol. Evol.* 30:2725–2729. doi:10.1093/molbev/mst197
- Van Eerdewegh, P., R.D. Little, J. Dupuis, R.G. Del Mastro, K. Falls et al. 2002. Association of the *ADAM33* gene with asthma and bronchial hyperresponsiveness. *Nature* 418:426–430. doi:10.1038/nature00878
- Varshney, R.K., R. Terauchi, and S.R. McCouch. 2014. Harvesting the promising fruits of genomics: Applying genome sequencing technologies to crop breeding. *PLoS Biol.* 12:E1001883. doi:10.1371/journal.pbio.1001883
- Velasco, R., A. Zharkikh, J. Affourtit, A. Dhingra, A. Cestaro et al. 2010. The genome of the domesticated apple (*Malus × domestica* Borkh.). *Nat. Genet.* 42:833–839. doi:10.1038/ng.654
- Watkins, C.B., J.F. Nocka, and B.D. Whitaker. 2000. Responses of early, mid and late season apple cultivars to postharvest application of 1-methylcyclopropene (1-MCP) under air and controlled atmosphere storage conditions. *Postharvest Biol. Technol.* 19:17–32. doi:10.1016/S0925-5214(00)00070-3
- Zhang, X.J., L.X. Wang, X.X. Chen, Y.L. Liu, R. Meng, Y.J. Wang, and Z.Y. Zhao. 2014. A and *MdMYB1* allele-specific markers controlling apple (*Malus × domestica* Borkh.) skin color and suitability for marker-assisted selection. *Genet. Mol. Res.* 13:9103–9114. doi:10.4238/2014.October.31.26
- Zhu, Y., K. Evans, and C. Peace. 2010. Utility testing of an apple skin color *MdMYB1* marker in two progenies. *Mol. Breed.* 27:525–532. doi:10.1007/s11032-010-9449-6